

WORK, ENERGY, AND POWER

Scalar Mechanics, Conservation Laws, and Variable Field
Systems

Subject: Classical Mechanics & Energy Vector Intersect Models

Level: Core Academic Reference Notes (Undergraduate Module)

Features: Fully Embedded Scalar Vector Geometries & Calculus Proofs

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1. MATHEMATICAL CONCEPT OF MECHANICAL WORK

In classical dynamics, **Work** is defined as the scalar quantitative measure of the energy transferred to or from a system via a force vector acting across a finite displacement path. It transitions vector force systems into a highly manageable scalar metric space.

1.1 Work Done by a Constant Force Field

When a constant external force vector $\vec{F}$ operates on a particle causing a linear displacement vector $\vec{d}$, the mechanical work done W is defined as the scalar inner dot product of those two vectors:

$$W = \vec{F} \cdot \vec{d} = F \cdot d \cdot \cos\theta$$

Where θ is the explicit spatial angle bounded between the positive orientation paths of the force and displacement vectors. Work is measured in Joules ($1 \text{ J} = 1 \text{ N} \cdot \text{m}$).

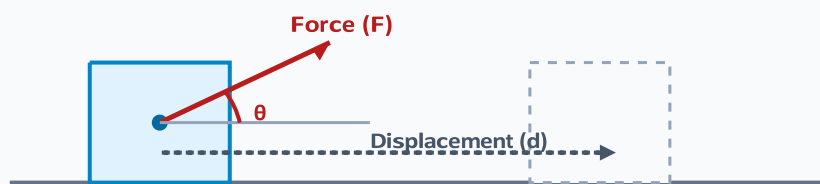


Figure 1.1: Vector Component Analysis of Work, indicating how the force vector projects an operational component ($F \cos \theta$) directly along the path of displacement.

1.2 Work Done by a Variable Force Field (Calculus Integration)

If a force field varies continuously over spatial coordinates, algebraic constant multiplications are invalid. We must evaluate an infinite summation of micro-displacements ($d\vec{r}$) using line integration parameters:

$$W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r} = \int_{x_1}^{x_2} F_x dx + \int_{y_1}^{y_2} F_y dy + \int_{z_1}^{z_2} F_z dz$$

2. KINETIC ENERGY AND THE WORK-ENERGY THEOREM

Kinetic Energy (\$K\$) represents the mathematical scalar capacity of a system to perform work strictly by virtue of its instantaneous operational state of translational velocity.

2.1 Rigorous Analytical Proof of the Work-Energy Theorem

The **Work-Energy Theorem** states that the integrated mechanical work performed by the net external force vector acting on a particle matches the net change in its Kinetic Energy configuration.

Calculus-Based Derivation Consider a variable force vector $\mathbf{F}$ accelerating an invariant mass m along a 1D coordinate axis x . The micro-differential work element is:

$$dW = F dx$$

Substituting Newton's Second Law ($F = m \frac{dv}{dt}$):

$$dW = m \frac{dv}{dt} dx$$

By shifting differential references via the calculus chain rule ($\frac{dx}{dt} = v$):

$$dW = m v dv$$

To find the total integrated work performed as velocity evolves from initial speed u to final speed v :

$$W = \int_u^v m v dv = m \cdot \left[\frac{1}{2} v^2 \right]_u^v$$

$$W = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$W = K_{\text{final}} - K_{\text{initial}} = \Delta K$$

3. CONSERVATIVE FORCES AND POTENTIAL ENERGY FIELD METRICS

Forces are structurally divided based on path integration traits into Conservative and Non-Conservative domains.

3.1 Mathematical Properties of Conservative Fields

A force field is strictly **Conservative** if the total closed-loop line integration around any arbitrary path returns perfectly zero:

$$\oint \vec{F} \cdot d\vec{r} = 0$$

Consequently, the work done in a conservative field (such as gravity or ideal springs) is completely independent of the path trajectory shape, relying exclusively on spatial endpoints. This independence allows us to define a scalar field function known as **Potential Energy (\$U\$)**:

$$\Delta U = U_f - U_i = -W_{\text{conservative}} = -\int_{r_i}^{r_f} \vec{F}_c \cdot d\vec{r}$$

3.2 Core Mechanical Field Profiles

1. **Gravitational Potential Energy Field:** Near the Earth's surface under constant downward field vector $\vec{g}$:

$$U(h) = mgh$$

2. **Elastic Spring Potential Energy Field (Hooke's Law):** For an ideal spring possessing a stiffness constant k , Hooke's restoring law states $\vec{F}_s = -k\vec{x}$. Integrating this profile yields:

$$W_s = \int_0^x (-kx) dx = -\frac{1}{2}kx^2 \quad U_{\text{spring}} = \frac{1}{2}kx^2$$

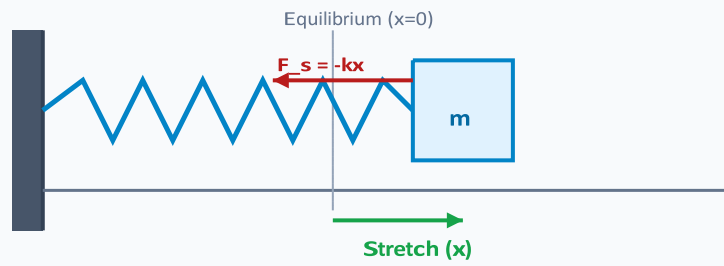


Figure 3.1: Hooke's Field Vector Arrangement, displaying how expanding a spring generates an inward restoring force field, storing elastic potential energy.

4. THE TOTAL CONSERVATION OF MECHANICAL ENERGY

When a system is isolated such that only conservative internal force fields act within its boundaries, the total macroscopic **Mechanical Energy (\$E\$)** of that system remains constant over time.

$$E = K + U \quad \Delta E = \Delta K + \Delta U = 0$$

This fundamental principle provides an efficient global scalar approach to solving complex, non-linear physical trajectories, eliminating the need to track localized instantaneous acceleration vectors.

5. ENGINEERING DYNAMICS OF POWER METRICS

Power (\$P\$) measures the operational time rate of executing mechanical work or transferring energy across system boundaries.

5.1 Instantaneous Scalar Velocity Multiplications

Average power is computed as total work divided by the overall elapsed time interval ($P_{\text{avg}} = \frac{\Delta W}{\Delta t}$). To evaluate the exact rate of energy transfer at a specific instant, we apply standard calculus derivatives:

$$P = dW / dt$$

Substituting the differential constant work relation expression ($dW = \vec{F} \cdot d\vec{r}$) yields:

$$P = d(\vec{F} \cdot d\vec{r}) / dt = \vec{F} \cdot (d\vec{r} / dt) = \vec{F} \cdot \vec{v}$$

Thus, the instantaneous mechanical power output equals the scalar dot product of the active force vector and the particle's instantaneous velocity vector. Power is measured in Watts ($1 \text{ W} = 1 \text{ J/s}$).

6. THEORETICAL MECHANICS OF MULTI-BODY COLLISIONS

Collisions represent short-duration internal structural force interactions between independent masses, modifying their global kinematic states based on momentum and energy laws.

6.1 Categorization and the Coefficient of Restitution

While total linear momentum is conserved across all isolated impacts, total kinetic energy may vary depending on the internal structural deformation properties of the colliding bodies:

- **Perfectly Elastic Collisions:** Total kinetic energy is perfectly preserved. No energy is lost to heat or sound ($\Delta K = 0$).
- **Inelastic Collisions:** Kinetic energy is partially lost to internal structural deformations, heat, or sound ($\Delta K \neq 0$).

To quantify these interaction variations, we define a scalar parameter known as the **Coefficient of Restitution** (e):

$$e = \text{Relative Speed of Separation} / \text{Relative Speed of Approach} = (v_2 - v_1) / (u_1 - u_2)$$

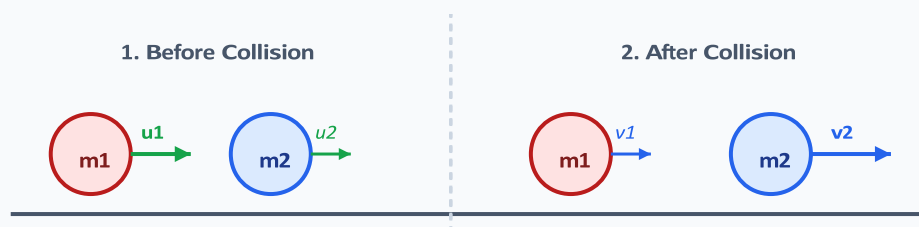


Figure 6.1: 1D Linear Collision Progression tracking initial relative approach parameters (u_1, u_2) transitioning into post-impact relative separation velocities (v_1, v_2).

Boundary Conditions of Restitution (e):

- For a perfectly elastic collision: $e = 1$
- For a standard inelastic collision: $0 < e < 1$
- For a perfectly inelastic collision (where objects stick together post-impact): $e = 0$